5.3 RL Agent Training

We train our RL agent for 5 million timesteps (96.5 hours). More details about the hyperparameters in Table 1.

Category	Hyperparameter	Value
PPO	Learning Rate	1e-4
	Batch Size	256
	Steps per Rollout (n_steps)	512
	Number of Epochs	15
	Clip Range (ϵ)	0.1
	Discount Factor (γ)	0.99
	GAE Lambda (λ)	0.98
	Entropy Coefficient (β_{ent})	$0.01 \rightarrow 0.0$
Training	Total Timesteps (T_{max})	5M
	Parallel Environments	32
	Biased Sampling Ratio (N_{cycle})	1
Reward	Fixed Key Budget (N_{budget})	5
	Preference Vectors ($\|\Omega\|$)	10
Exploration	Envelope Coefficient (λ_{env})	0.0
	KL Coefficient (λ_{KL})	0.0
Constraint	Training Budgets (B)	[230, 236, 369, 9M]
	Constraint Method	Lag-PID + NATO-SC
	PID Gains (K_p, K_i, K_d)	(0.005, 0.001, 0.001)
	Dual Multiplier Range (λ)	[0.0, 20.0]
	NATO-SC Penalty Coeff.	50.0
Network	Budget Conditioning	FiLM
	Policy (Rule Head) Hidden Layers	[64, 32]
	Policy (Position Head) Hidden Layers	[32, 32]
	Value Network Hidden Layers	[128, 64, 32]

Table 1. Hyperparameter settings for the unified CMORL agent under noise constraint.

5.4 Main Results

5.4.1 Unified CMO-RL vs. Original CHEHAB RL . We compare CHEHAB with the original CHEHAB RL [**?**]. CHEHAB RL optimizes execution time only and ignores both rotation keys and noise. To give it a way to reduce key memory, we enable its rotation key selection pass for this experiment only. This pass runs after the RL agent and decomposes rotations into power-of-two rotations to reduce the number of keys (Sec. 5.2.2). We fix the noise budget to $B = 230$ bits, a budget that CHEHAB RL exceeds on most benchmarks. We fix the preference to $\vec{w} = [0.9, 0.1]^\top$, which co-optimizes speed and key memory. We report the raw output of the agent, without safety rollback (Sec. 4.3.1).

Figure 2 and Table 2 show the results. CHEHAB RL exceeds the budget on 10 of the 14 benchmarks, by up to 17.8% (Hamm. Dist. 32). CHEHAB satisfies the budget on all benchmarks. The constraint does not cost execution time: CHEHAB is as fast or faster on 13 benchmarks, with a geometric-mean speedup of 1.02×. It also needs fewer keys: 44 keys (792 MB) in total, against 64 (1152 MB) for CHEHAB RL, a reduction of 31%. We attribute

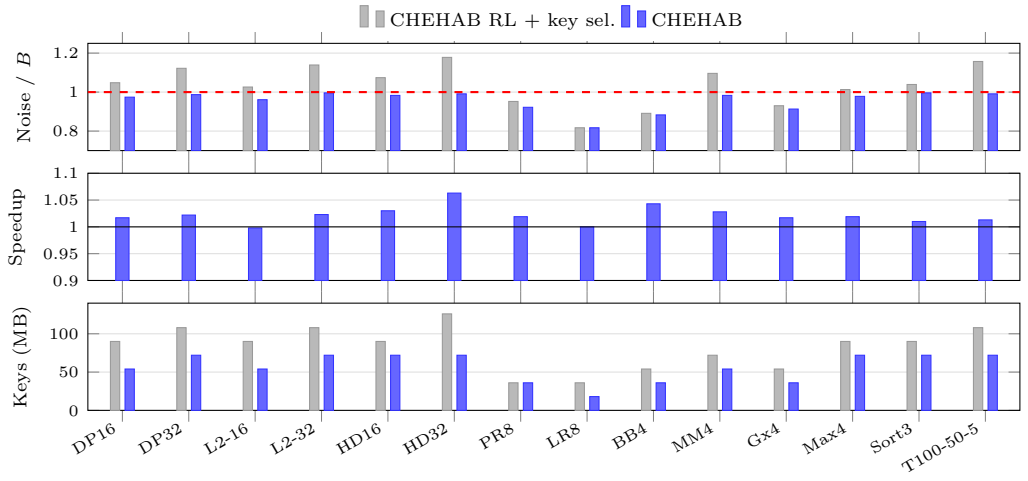


Fig. 2. CHEHAB vs. CHEHAB RL with rotation key selection ($B = 230$ bits, $\vec{w} = [0.9, 0.1]^\top$). Top: noise normalized by B (bars above the dashed line violate the budget). Middle: speedup of CHEHAB (higher is better). Bottom: rotation key size (lower is better). Placeholder data.

Table 2. Noise (bits), budget violation, execution time (ms), and number of rotation keys ($B = 230$ bits, $\vec{w} = [0.9, 0.1]^\top$). Placeholder data.

	CHEHAB RL + key sel.				CHEHAB			
Benchmark	Noise	OK	Time	Keys	Noise	OK	Time	Keys
Dot Product 16	241	×	412	5	224	✓	405	3
Dot Product 32	258	×	655	6	227	✓	641	4
L2 Distance 16	236	×	430	5	221	✓	431	3
L2 Distance 32	262	×	790	6	229	✓	772	4
Hamm. Dist. 16	247	×	980	5	226	✓	951	4
Hamm. Dist. 32	271	×	1490	7	228	✓	1402	4
Poly. Reg. 8	219	✓	315	2	212	✓	309	2
Linear Reg. 8	188	✓	140	2	188	✓	140	1
Box Blur 4	205	✓	96	3	203	✓	92	2
Mat. Mul. 4	252	×	512	4	226	✓	498	3
Gx 4	214	✓	530	3	210	✓	521	2
Max 4	233	×	1210	5	225	✓	1188	4
Sort 3	239	×	1380	5	229	✓	1366	4
tree-100-50-5	266	×	2010	6	228	✓	1985	4
Violations			10/14				0/14	

this difference to when keys are handled. CHEHAB RL fixes the vectorization first and then replaces rotations by chains of rotations, which adds operations and noise. CHEHAB accounts for keys and noise while it chooses the vectorization.

5.4.2 Joint Optimization vs. Separate Optimization .

5.4.3 Trade-off Evolution Across Preference Weights .

5.4.4 Multi-Objective Optimization Quality Across Benchmarks .

5.4.5 Constraint Satisfaction Under Different Noise Budgets .

5.4.6 Noise Estimator Accuracy .